

Motinath R

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EDUCATION

Vellore Institute of Technology – AP	2022 - 2026
<i>B. Tech, Electronics and Communication Engineering</i> CGPA: 7.92/10	<i>Andhra Pradesh, India</i>

WORK EXPERIENCE

Silicofeller Quantum Engineer	May 2026
Developed core AI capabilities for a next-generation quantum Electronic Design Automation (EDA) platform, enabling automated superconducting quantum chip design and validation.	
New Technologies Institution Full Stack Developer Intern	Dec 2023
Collaborated with the development team to translate product requirements into structured UI specifications, supporting sprint delivery, Built and tested UI components	

PROJECTS

7T/8T SRAM Design and Analysis	Mar 2026
<i>Tools: Cadence Virtuoso CMOS Design Analog Simulation SNM Analysis</i>	
Designed and simulated 7T/8T SRAM in Cadence with asymmetrical feedback to improve write ability and stability, validated through SNM and DC analysis.	
Physio Trust – Health Signal Reliability System	Dec 2025
<i>Tools: Python Signal Processing Embedded-ready Pipeline</i>	
- Built a signal validation pipeline to detect unreliable physiological readings caused by motion artifacts and electrical noise in wearable health sensors. - Applied contextual behaviour analysis (activity state + signal morphology) to filter false-positive readings and improve downstream monitoring reliability.	
Adaptive Vehicle Navigation Using ANFIS and ROS	Nov 2025
<i>Tools: ROS2 Gazebo Python ANFIS Controller</i>	
- Engineered an ANFIS-based intelligent controller for autonomous obstacle avoidance and path planning in ROS2. - Validated navigation across multiple simulation scenarios in Gazebo with real-time sensing and decision-making integration	
Contentlyzer - SEO Audit Tool	Jul 2025
<i>Tools: React.js Google Search Console API REST</i>	
Built a web dashboard automating website content analysis and SEO auditing — reduced manual audit effort by 50% across test deployments.	
Electric Vehicle Dynamics Simulation	Apr 2025
<i>Tools: MATLAB Simulink BLDC Motor Model Li-ion BMS</i>	
Simulated a full EV scooter system including BLDC motor and battery management in MATLAB Simulink, achieving 90% operational efficiency across varying load conditions	

SKILLS SUMMARY

- Programming Languages:** Python (Basic), Java, Embedded C (Basic)
- Simulation:** MATLAB Simulink, IBM Qiskit Metal, AutoCAD (Basic), Cadence Virtuoso, Gazebo,
- Platforms & Tools:** ROS2, Arduino IDE, Google Search Console, Figma, Postman, Git, Canva, Linux (Basic)

CERTIFICATIONS

Oracle Cloud Infrastructure 2025 Generative AI Professional	Aug 2025
Oracle Fusion Cloud Applications ERP Process Essentials Certified – Rel 1	May 2025
India Semiconductor Workforce Development Program - IISc	Jun 2025
Foundation Course on Embedded System Design - Maven Silicon	Jun 2024